Differential Splicing with Subisoform Covariance

Overview

Questions and unit of analysis

The analysis asks two distinct questions:

1. Does local splice-path usage differ by experimental condition within a cell type?
2. Does local splice-path usage differ across cell types within the same mice?

The primary analysis aggregates cells into *mouse-by-condition-by-cell-type pseudobulks*. An evaluated alternative retains each cell's EC counts and isoform baseline in the local likelihood but estimates one shared path effect for the pseudobulk. Both approaches keep biological replication at the mouse level; cells are never treated as independent condition replicates.

A splice block is a local region between constitutive exonic anchors. Transcripts that take the same path through that region are collapsed to one subisoform. This targets local splicing choices, which can be identifiable even when the corresponding full-length isoforms are not.

For condition effects, each hypothesis is indexed by a splice block and a cell type. It is an omnibus test of

$$H_0: \quad \text{mean path usage is equal across the included conditions.}$$

Only conditions represented by at least three mouse pseudobulks are included. With two conditions and two paths, this is a precision-weighted differential-PSI test. With more conditions or paths, all condition coefficients and path log-ratios are tested jointly.

For cell-type effects, each hypothesis is indexed only by a nonredundant splice-path partition. It is an omnibus test of equal path usage across the included cell types, after absorbing mouse and condition into a mouse-specific fixed effect. Cell types must be represented in at least three mice, and the primary FDR family requires at least 30 pseudobulk observations per partition. These two hypothesis families are reported separately.

Workflow

The analysis proceeds in five steps:

1. Construct local splice blocks from the isoform annotation and collapse isoforms to paths within each block.
2. Aggregate paired oligo(dT) and random-hexamer EC counts, and hierarchical isoform estimates, within each mouse, condition, and cell type.
3. Refine path usage in each pseudobulk with the exact paired-primer EC likelihood. Optionally, retain separate cell likelihood terms while sharing one pseudobulk path perturbation.

4. Estimate path-logratio covariance from EC ambiguity and UMI sampling. Estimates with weak information or boundary path usage are retained in output but excluded from Gaussian testing.

5. Fit mouse-replicate GLS condition models and compositional cell-type models. Test condition within cell type and cell type within mouse as separate hypothesis families.

Two inferential covariance calculations are reported. The *conditional* calculation fixes the fitted isoform mixture within each path. The *profile* calculation allows all isoform directions to vary as nuisance parameters. They are sensitivity analyses for the same block-by-cell-type hypothesis, not independent biological hypotheses.

Notation and Dimensions

An *equivalence class* (EC) is a set of isoforms compatible with one UMI. An EC count is therefore observed, whereas its allocation among compatible isoforms is latent. A *splice path* is an ordered local exon chain between two constitutive anchors. A *subisoform* is the set of full-length isoforms taking the same path through one block. PSI means percent spliced in; for two paths it is one component of the path-composition vector. ILR means isometric log-ratio, an orthonormal coordinate system for a composition.

Let $g = g(b)$ denote the gene containing block b, and let $\mathcal{I}_g$ be the set of isoforms retained by the paired-primer model and assigned to gene g by the isoform-to-gene table. Define $T_g = |\mathcal{I}_g| \in \mathbb{N}$. This is a gene-level set, not the set of isoforms whose exons overlap the block, and it is therefore shared by every block of the gene. Let $\mathcal{I}_b^{\mathrm{sp}} \subseteq \mathcal{I}_g$ be the retained, annotated spliced isoforms represented in the block's GTF-derived path map. Isoforms in $\mathcal{I}_g \setminus \mathcal{I}_b^{\mathrm{sp}}$, including modeled unspliced precursors, remain nuisance isoforms in the EC likelihood.

Let $n_b, S_b \in \mathbb{N}$ be the numbers of qualifying pseudobulks and represented spliced paths for block b, with $S_b \geq 2$. Then $d_b = S_b - 1 \in \mathbb{N}$ is its number of independent log-ratios. There are $P = 2$ primer protocols. Primer p has $E_{bp} \in \mathbb{N}$ ECs for gene $g(b)$. Pseudobulk i contains $H_{bi} \in \mathbb{N}$ cells. The principal symbols are:

symbol	dimension or domain	meaning
b	$\{1, \ldots, \mathcal{B}\}$	splice-block index
i	$\{1, \ldots, n_b\}$	pseudobulk index within block b
p	$\{1, \ldots, P\}$	primer-protocol index
h	$\{1, \ldots, H_{bi}\}$	cell index within pseudobulk i
t	$\{1, \ldots, T_g\}$	isoform index within gene $g(b)$
s	$\{1, \ldots, S_b\}$	splice-path index
$\boldsymbol{c}_{bip}$	$\mathbb{N}_0^{E_{bp}}$	pseudobulk EC counts
A_{bp}	$\mathbb{R}_+^{E_{bp} \times T_g}$	weighted EC-by-isoform design
$\boldsymbol{\theta}_{bi}$	$\mathbb{R}_+^{T_g}$	isoform abundance
C_b	$\{0,1\}^{S_b \times T_g}$	isoform-to-path map
Q_b	$\mathbb{R}^{S_b \times d_b}$	Helmert ILR basis
$\boldsymbol{\delta}_{bi}$	$\mathbb{R}^{d_b}$	local path perturbation
$\boldsymbol{\psi}_{bi}$	$\Delta^{S_b - 1}$	path composition
$\boldsymbol{y}_{bi}$	$\mathbb{R}^{d_b}$	ILR response
V_{bi}	$\mathbb{S}_+^{d_b}$	ILR inferential covariance

Here $\mathcal{B} \in \mathbb{N}$ is the number of blocks, $\mathbb{N}_0 = \{0, 1, \ldots\}$, $\Delta^{S_b-1} = \{\boldsymbol{v} \in \mathbb{R}_+^{S_b} : \mathbf{1}'\boldsymbol{v} = 1\}$, and $\mathbb{S}_+^{d_b}$ is the set of symmetric positive-semidefinite $d_b \times d_b$ matrices. Bold lower-case symbols are vectors, upper-case symbols are matrices, $\mathbf{1}_k \in \mathbb{R}^k$ is a vector of ones, $\mathbf{0}_k \in \mathbb{R}^k$ is a vector of zeros, $I_k \in \mathbb{R}^{k\times k}$ is the identity, $\odot$ is elementwise multiplication, and $(\cdot)^+$ is the Moore–Penrose pseudoinverse.

Helmert basis and ILR coordinates

The simplex has only $d_b = S_b - 1$ free directions because path proportions sum to one. The Helmert matrix Q_b is a fixed orthonormal basis for the sum-zero subspace:

$$Q_b' Q_b = I_{d_b}, \qquad Q_b' \mathbf{1}_{S_b} = \mathbf{0}_{d_b}.$$

Using one-based row $r \in \{1, \ldots, S_b\}$ and column $j \in \{1, \ldots, d_b\}$, its entries are

$$(Q_b)_{rj} = \begin{cases} 1/\sqrt{j(j+1)}, & r \le j, \\ -j/\sqrt{j(j+1)}, & r = j+1, \\ 0, & r > j+1. \end{cases}$$

For any strictly positive $\boldsymbol{\psi}_{bi} \in \Delta^{S_b-1}$, the ILR coordinate is $\boldsymbol{y}_{bi} = Q_b' \log \boldsymbol{\psi}_{bi} \in \mathbb{R}^{d_b}$, where the logarithm is elementwise. For $S_b = 2$, $Q_b = (1, -1)'/\sqrt{2}$ and $y_{bi} = \log(\psi_{bi1}/\psi_{bi2})/\sqrt{2}$. Thus a two-path ILR test is a scaled log-odds, or differential-PSI, test.

Statistical Model

Paired-primer EC likelihood

For block b, pseudobulk i, and primer p, define $N_{bip} = \mathbf{1}'_{E_{bp}} \boldsymbol{c}_{bip} \in \mathbb{N}_0$, $\boldsymbol{q}_{bip} = A_{bp} \boldsymbol{\theta}_{bi} \in \mathbb{R}_+^{E_{bp}}$, and $Z_{bip} = \mathbf{1}'_{E_{bp}} \boldsymbol{q}_{bip} \in \mathbb{R}_+$. Conditional on the primer-specific UMI total,

$$\boldsymbol{c}_{bip} \mid N_{bip}, \boldsymbol{\theta}_{bi} \sim \text{Multinomial}\left(N_{bip}, \boldsymbol{q}_{bip}/Z_{bip}\right).$$

The oligo(dT) and random-hexamer protocols use separate A_{bp} matrices and likelihood terms. Their different isoform exposures therefore model primer-specific positional coverage without forcing a common bias profile.

The genome-wide hierarchical model supplies $\boldsymbol{\theta}_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$. Let $s_b(t) \in \{1, \ldots, S_b\}$ map isoform $t \in \mathcal{I}_b^{\text{sp}}$ to its path. The baseline spliced-isoform mass is the scalar $\omega_{bi}^{\text{sp}} = \sum_{t \in \mathcal{I}_b^{\text{sp}}} \theta_{bit}^{(0)} \in \mathbb{R}_+$. For $t \in \mathcal{I}_b^{\text{sp}}$, local refinement uses

$$\theta_{bit}(\boldsymbol{\delta}_{bi}) = \omega_{bi}^{\text{sp}} \frac{\theta_{bit}^{(0)} \exp\{(Q_b)_{s_b(t),\cdot} \boldsymbol{\delta}_{bi}\}}{\sum_{u \in \mathcal{I}_b^{\text{sp}}} \theta_{biu}^{(0)} \exp\{(Q_b)_{s_b(u),\cdot} \boldsymbol{\delta}_{bi}\}},$$

where $(Q_b)_{s,\cdot} \in \mathbb{R}^{1\times d_b}$. Nuisance-isoform abundances are unchanged. This preserves total spliced mass and the relative isoform mixture within each path while estimating only $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$.

Path response and inferential covariance

Each column of $C_b \in \{0, 1\}^{S_b \times T_g}$ corresponding to $t \in \mathcal{I}_b^{\text{sp}}$ contains one 1 in row $s_b(t)$; nuisance isoform columns are zero. Define

$$\boldsymbol{a}_{bi} = C_b \boldsymbol{\theta}_{bi} \in \mathbb{R}_+^{S_b}, \qquad \boldsymbol{\psi}_{bi} = \frac{\boldsymbol{a}_{bi}}{\mathbf{1}'_{S_b} \boldsymbol{a}_{bi}} \in \Delta^{S_b-1}, \qquad \boldsymbol{y}_{bi} = Q_b' \log \boldsymbol{\psi}_{bi} \in \mathbb{R}^{d_b}.$$

The perturbation parameterization implies $\boldsymbol{y}_{bi} = \boldsymbol{y}_{bi}^{(0)} + \boldsymbol{\delta}_{bi}$, where $\boldsymbol{y}_{bi}^{(0)} \in \mathbb{R}^{d_b}$ is computed from the baseline.

Let $I_{\delta,bi} \in \mathbb{S}_+^{d_b}$ be the expected Fisher information of the summed paired-primer log likelihood with respect to $\boldsymbol{\delta}_{bi}$. The conditional covariance is $V_{bi}^{\text{cond}} = I_{\delta,bi}^{-1} \in \mathbb{S}_+^{d_b}$ when the information is positive definite. It conditions on the fitted total spliced mass, the isoform mixture within each path, and every nuisance-isoform abundance. Its only free directions are the d_b path perturbations.

Full-isoform nuisance (profile) covariance

The profile calculation is a covariance sensitivity analysis, not a second estimate of path usage. It retains the same $\widehat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{T_g}$ obtained from the constrained local fit, but asks how uncertain its path ILRs would be if all T_g isoform log-abundances were locally free. Equivalently, after a smooth reparameterization into path ILRs and $T_g - d_b$ nuisance coordinates, it approximates the path-ILR block of the inverse Fisher information after profiling the nuisance coordinates. No unconstrained isoform likelihood is optimized, so *profile* below means this local expected-Fisher approximation rather than an exact profile-likelihood fit.

Set $\boldsymbol{\eta}_{bi} = \log \widehat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}^{T_g}$, $D_{bi} = \operatorname{diag}(\widehat{\boldsymbol{\theta}}_{bi}) \in \mathbb{R}^{T_g \times T_g}$, $\boldsymbol{d}_{bp} = A_{bp}' \mathbf{1}_{E_{bp}} \in \mathbb{R}^{T_g}$, and $\widehat{\boldsymbol{q}}_{bip} = A_{bp} \widehat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{E_{bp}}$, $\widehat{Z}_{bip} = \mathbf{1}_{E_{bp}}' \widehat{\boldsymbol{q}}_{bip} \in \mathbb{R}_+$, and $\boldsymbol{r}_{0,bip} = (\boldsymbol{d}_{bp} \odot \widehat{\boldsymbol{\theta}}_{bi}) / \widehat{Z}_{bip} \in \mathbb{R}^{T_g}$. The primer-specific expected information is

$$I_{\eta,bip} = N_{bip} \left[\frac{D_{bi} A_{bp}' \operatorname{diag}(\widehat{\boldsymbol{q}}_{bip}^{-1}) A_{bp} D_{bi}}{\widehat{Z}_{bip}} - \boldsymbol{r}_{0,bip} \boldsymbol{r}_{0,bip}' \right] \in \mathbb{S}_+^{T_g}.$$

The paired-primer information is $I_{\eta,bi} = \sum_{p=1}^{P} I_{\eta,bip} \in \mathbb{S}_+^{T_g}$.

The covariance must be propagated through the isoform-to-path map. Let $\omega_{bi}^{\text{sp}} = \sum_{t \in \mathcal{I}_b^{\text{sp}}} \widehat{\theta}_{bit} \in \mathbb{R}_+$ and $a_{bis} = \sum_{t: s_b(t) = s} \widehat{\theta}_{bit} \in \mathbb{R}_+$. Define $R_{bi} = \partial \log \boldsymbol{\psi}_{bi} / \partial \boldsymbol{\eta}_{bi}' \in \mathbb{R}^{S_b \times T_g}$ entrywise by

$$(R_{bi})_{st} = \begin{cases} \mathbb{1}\{s_b(t) = s\} \widehat{\theta}_{bit} / a_{bis} - \widehat{\theta}_{bit} / \omega_{bi}^{\text{sp}}, & t \in \mathcal{I}_b^{\text{sp}}, \\ 0, & t \notin \mathcal{I}_b^{\text{sp}}. \end{cases}$$

Thus

$$J_{bi} = \frac{\partial \boldsymbol{y}_{bi}}{\partial \boldsymbol{\eta}_{bi}'} = Q_b' R_{bi} \in \mathbb{R}^{d_b \times T_g}.$$

The multinomial likelihood necessarily has a global-scale null direction, and EC ambiguity can create additional null directions. Write the symmetric eigendecomposition

$$I_{\eta,bi} = U_{bi} \operatorname{diag}(\boldsymbol{\lambda}_{bi}) U_{bi}',$$

where $U_{bi} \in \mathbb{R}^{T_g \times T_g}$ is orthogonal and $\boldsymbol{\lambda}_{bi} \in \mathbb{R}^{T_g}$. An eigenvalue is treated as positive when $\lambda_{bit} > 10^{-8} \max_u |\lambda_{biu}|$. Let $\mathcal{P}_{bi} \subseteq \{1, \ldots, T_g\}$ index these positive eigenvalues, let $\mathcal{N}_{bi}$ index the remainder, and let $U_{bi,+} \in \mathbb{R}^{T_g \times |\mathcal{P}_{bi}|}$ and $U_{bi,0} \in \mathbb{R}^{T_g \times |\mathcal{N}_{bi}|}$ contain the corresponding eigenvectors.

The path response is identifiable only if its Jacobian annihilates every information-null direction. The implementation requires

$$\|J_{bi} U_{bi,0}\|_F \le 10^{-9} \max\{\|J_{bi}\|_F, 1\}.$$

If this fails, V_{bi}^{prof} is set to an infinite matrix and the estimate is excluded from profile testing. Otherwise,

$$V_{bi}^{\text{prof}} = J_{bi} U_{bi,+} \operatorname{diag}(\boldsymbol{\lambda}_{bi,\mathcal{P}}^{-1}) U_{bi,+}' J_{bi}' = J_{bi} I_{\eta,bi}^{+} J_{bi}' \in \mathbb{S}_+^{d_b}.$$

Unlike V_{bi}^{cond}, this covariance allows EC-supported changes in within-path isoform mixtures, spliced mass, and nuisance isoforms to absorb information that would otherwise identify a path contrast. It is therefore the more conservative sensitivity calculation when those nuisance directions are weakly identified.

Cell-resolved likelihood

For cell h in pseudobulk i, let $\boldsymbol{c}_{bihp} \in \mathbb{N}_0^{E_{bp}}$, $N_{bihp} = \mathbf{1}'_{E_{bp}} \boldsymbol{c}_{bihp} \in \mathbb{N}_0$, and $\boldsymbol{\theta}_{bih}^{(0)} \in \mathbb{R}_+^{T_g}$ be its EC counts, total, and baseline. Every cell shares the same pseudobulk perturbation $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$, but its isoform vector $\boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi}) \in \mathbb{R}_+^{T_g}$ is normalized using its own baseline. The objective is the scalar

$$\ell_{bi}(\boldsymbol{\delta}_{bi}) = \sum_{h=1}^{H_{bi}} \sum_{p=1}^{P} \log \text{Multinomial} \left(\boldsymbol{c}_{bihp}; N_{bihp}, \frac{A_{bp}\boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})}{\mathbf{1}'_{E_{bp}} A_{bp}\boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})} \right).$$

Thus cells contribute assignment information but do not become biological replicates. Let $U_{bih} \in \mathbb{R}_+$ be the cell's total UMI count and $g_{bih} \in [0,1]$ its decoded proportion for the block's gene. The scalar weight is $w_{bih} = U_{bih} g_{bih}$, and the reported pseudobulk composition is

$$\boldsymbol{\psi}_{bi} = \frac{\sum_{h=1}^{H_{bi}} w_{bih} \boldsymbol{\psi}_{bih}}{\sum_{h=1}^{H_{bi}} w_{bih}} \in \Delta^{S_b - 1}.$$

Cell- and primer-specific Fisher information matrices in $\mathbb{S}_+^{d_b}$ are summed, inverted, and propagated through this weighted average to obtain $V_{bi} \in \mathbb{S}_+^{d_b}$.

Effective counts and compositional regression

Treating all gene UMIs as known path assignments is overconfident when ECs are ambiguous. For each fitted $\boldsymbol{\psi}_{bi}$, define the unit-multinomial ILR covariance

$$G_{bi} = Q_b' \operatorname{diag}(\boldsymbol{\psi}_{bi}^{-1}) Q_b \in \mathbb{S}_+^{d_b}.$$

The scalar effective count matches G_{bi}/n to the EC-derived V_{bi}^{cond} in Frobenius inner product $\langle U, W \rangle_F = \operatorname{tr}(U'W)$, for $U, W \in \mathbb{R}^{d_b \times d_b}$:

$$n_{bi}^{\text{eff}} = \min \left\{ N_{bi}^{\text{gene}}, \frac{\langle G_{bi}, G_{bi} \rangle_F}{\langle G_{bi}, V_{bi}^{\text{cond}} \rangle_F} \right\} \in \mathbb{R}_+,$$

where $N_{bi}^{\text{gene}} \in \mathbb{N}_0$ is the gene UMI count. Fractional path counts are $\boldsymbol{x}_{bi} = n_{bi}^{\text{eff}} \boldsymbol{\psi}_{bi} \in \mathbb{R}_+^{S_b}$, with $\mathbf{1}'_{S_b} \boldsymbol{x}_{bi} = n_{bi}^{\text{eff}}$.

For one block, stack $n = n_b \in \mathbb{N}$ observations in a design $X \in \mathbb{R}^{n \times k}$, where $k \in \mathbb{N}$, and coefficient matrix $\boldsymbol{B}_b \in \mathbb{R}^{k \times d_b}$. The mean composition $\boldsymbol{\mu}_{bi} \in \Delta^{S_b - 1}$ is defined by reference-path logits

$$\log \frac{\mu_{bis}}{\mu_{biS_b}} = (X\boldsymbol{B}_b)_{is}, \qquad s = 1, \ldots, d_b.$$

The generalized Dirichlet–multinomial likelihood, extended to fractional counts through gamma functions, is

$$\boldsymbol{x}_{bi} \mid n_{bi}^{\text{eff}} \sim \text{DirichletMultinomial} \left(n_{bi}^{\text{eff}}, \kappa_b \boldsymbol{\mu}_{bi} \right),$$

where $\kappa_b \in \mathbb{R}_+ \cup \{\infty\}$ is a block-level concentration. Larger κ_b means less overdispersion; the endpoint $\kappa_b = \infty$ is exactly multinomial.

For nested designs $X_0 \in \mathbb{R}^{n\times k_0}$ and $X_1 \in \mathbb{R}^{n\times k_1}$, with $k_1 \geq k_0$ and the first $k_0, k_1 \in \mathbb{N}$, and the first k_0 columns of X_1 equal to X_0, let $\widehat{\ell}_0, \widehat{\ell}_1 \in \mathbb{R}$ be maximized scalar log likelihoods. The statistic and nominal degrees of freedom are

$$\Lambda_b = 2(\widehat{\ell}_1 - \widehat{\ell}_0) \in \mathbb{R}_+, \qquad \nu_b = (k_1 - k_0)d_b \in \mathbb{N}_0.$$

The fixed-concentration analysis estimates $\widehat{\kappa}_{0b} \in \mathbb{R}_+ \cup \{\infty\}$ under X_0 and uses it under X_1. The free-concentration sensitivity estimates $\widehat{\kappa}_{1b} \in \mathbb{R}_+ \cup \{\infty\}$ under X_1. In the latter case both models contain one concentration nuisance parameter, so ν_b is unchanged. Both finite and exact multinomial endpoints are evaluated.

Gaussian mouse-replicate regression

For a condition test within one block and cell type, stack $\widehat{\boldsymbol{y}}_{bi} \in \mathbb{R}^{d_b}$ over $n \in \mathbb{N}$ mice. Let $X_b \in \mathbb{R}^{n\times k}$, $k \in \mathbb{N}$, contain an intercept and condition indicators, $\boldsymbol{B}_b^{\mathrm{GLS}} \in \mathbb{R}^{k\times d_b}$ be regression coefficients, and $\boldsymbol{\epsilon}_{bi} \in \mathbb{R}^{d_b}$ be residuals:

$$\widehat{\boldsymbol{y}}_{bi} = (X_{b,i\cdot}\boldsymbol{B}_b^{\mathrm{GLS}})' + \boldsymbol{\epsilon}_{bi}, \qquad \mathrm{Var}(\boldsymbol{\epsilon}_{bi}) = V_{bi} + \tau_b^2 I_{d_b} \in \mathbb{S}_+^{d_b}.$$

Here $X_{b,i\cdot} \in \mathbb{R}^{1\times k}$, and $\tau_b^2 \in \mathbb{R}_+$ is residual between-mouse variance estimated by REML under the null. If $r_b \in \mathbb{N}$ non-reference condition columns are tested, their coefficients form $\boldsymbol{\beta}_b^{\mathrm{test}} \in \mathbb{R}^{r_b d_b}$ with covariance $\Sigma_b^{\mathrm{test}} \in \mathbb{S}_+^{r_b d_b}$. The Wald statistic is

$$W_b = (\boldsymbol{\beta}_b^{\mathrm{test}})'(\Sigma_b^{\mathrm{test}})^{-1}\boldsymbol{\beta}_b^{\mathrm{test}} \in \mathbb{R}_+, \qquad \nu_b^{\mathrm{GLS}} = r_b d_b.$$

Paired cell-type test and permutations

For a cell-type test, let $M_b, L_b \in \mathbb{N}$ be the numbers of represented biological mice and cell types, respectively. The null design $X_{0b} \in \{0,1\}^{n_b\times M_b}$ contains an intercept-coded set of mouse fixed effects. The alternative $X_{1b} \in \mathbb{R}^{n_b\times(M_b+L_b-1)}$ appends treatment-coded cell-type indicators. Consequently $k_0 = M_b$, $k_1 = M_b + L_b - 1$, and $\nu_b = (L_b - 1)d_b$. The tested coefficient block is $\boldsymbol{B}_b^{\mathrm{cell}} \in \mathbb{R}^{(L_b-1)\times d_b}$, and

$$H_{0b} : \boldsymbol{B}_b^{\mathrm{cell}} = 0 \quad \text{versus} \quad H_{1b} : \boldsymbol{B}_b^{\mathrm{cell}} \neq 0.$$

Mouse effects absorb condition and every shift shared by cell types from the same mouse.

For $R = 500 \in \mathbb{N}$ permutations and $r \in \{1, \ldots, R\}$, cell-type labels are shuffled only among rows from the same mouse, producing $X_{1b}^{(r)} \in \mathbb{R}^{n_b\times(M_b+L_b-1)}$ and statistic $\Lambda_b^{(r)} \in \mathbb{R}_+$. Let $\varepsilon_b = 10^{-6}\max(1, |\Lambda_b|) \in \mathbb{R}_+$. The finite-permutation p-value is

$$p_b = \frac{1 + \sum_{r=1}^{R} \mathbb{1}\{\Lambda_b^{(r)} \geq \Lambda_b - \varepsilon_b\}}{R+1} \in (0,1], \qquad R = 500.$$

The null coefficients and $\widehat{\kappa}_{0b}$ are reused, but each permuted alternative is optimized. The tolerance counts numerically equal likelihood optima as ties. Benjamini–Hochberg correction is applied across canonical partitions with at least 30 pseudobulk rows.

Joint condition tests

ACAT combines $J \in \mathbb{N}$ component p-values $p_j \in (0,1)$, $j = 1, \ldots, J$, into the scalar

$$p_{\mathrm{ACAT}} = \frac{1}{2} - \frac{1}{\pi}\arctan\left[\frac{1}{J}\sum_{j=1}^{J}\tan\{\pi(1/2 - p_j)\}\right] \in (0,1).$$

One level combines cell-type-specific p-values for a path partition; a second combines distinct partitions within a gene.

The shared-effect Gaussian model for block b uses $M \in \mathbb{N}$ mice indexed by $m \in \{1, \ldots, M\}$, and $L_m \in \mathbb{N}$ cell types for mouse m, indexed by $c \in \{1, \ldots, L_m\}$. Its response is $\boldsymbol{y}_{bmc} \in \mathbb{R}^{d_b}$, design row $X_{bmc} \in \mathbb{R}^{1\times k}$, where $k \in \mathbb{N}$, coefficient matrix $\boldsymbol{B}_b \in \mathbb{R}^{k\times d_b}$, mouse random effect $\boldsymbol{u}_{bm} \in \mathbb{R}^{d_b}$, and residual $\boldsymbol{e}_{bmc} \in \mathbb{R}^{d_b}$:

$$\boldsymbol{y}_{bmc} = (X_{bmc}\boldsymbol{B}_b)' + \boldsymbol{u}_{bm} + \boldsymbol{e}_{bmc}.$$

The random terms satisfy $\mathrm{E}(\boldsymbol{u}_{bm}) = \mathrm{E}(\boldsymbol{e}_{bmc}) = \boldsymbol{0}_{d_b}$, $\mathrm{Var}(\boldsymbol{u}_{bm}) = \tau^2_{\text{mouse}} I_{d_b}$, and $\mathrm{Var}(\boldsymbol{e}_{bmc}) = V_{bmc} + \tau^2_{\text{resid}} I_{d_b}$. They are independent across mice, and residuals are independent across cell types conditional on $\boldsymbol{u}_{bm}$. Hence, for $c \neq c'$,

$$\mathrm{Cov}(\boldsymbol{y}_{bmc}, \boldsymbol{y}_{bmc'}) = \tau^2_{\text{mouse}} I_{d_b}, \qquad \mathrm{Var}(\boldsymbol{y}_{bmc}) = V_{bmc} + (\tau^2_{\text{mouse}} + \tau^2_{\text{resid}}) I_{d_b},$$

where $V_{bmc} \in \mathbb{S}^{d_b}_{+}$, and $\tau^2_{\text{mouse}}, \tau^2_{\text{resid}} \in \mathbb{R}_+$ are shared-mouse and residual variance components estimated by two-component REML under the no-condition model. Condition labels are permuted once per mouse and applied to all its cell types.

Multinomial-logit GLMM

The formal generalized linear mixed model uses the same shared-condition design as the joint Gaussian model but retains the multinomial response. Let $\widetilde{\boldsymbol{x}}_{bmc} \in \mathbb{N}_0^{S_b}$ be integer path counts with $\widetilde{n}_{bmc} = \boldsymbol{1}'_{S_b}\widetilde{\boldsymbol{x}}_{bmc} \in \mathbb{N}$. They are obtained from the fractional effective counts $\boldsymbol{x}_{bmc}$ by rounding n^{eff}_{bmc} to the nearest positive integer and distributing the remaining molecules to paths by largest fractional remainder. This makes the conditional response a proper multinomial while changing each effective total by at most one.

For design row $X_{bmc} \in \mathbb{R}^{1\times k}$, fixed coefficients $\boldsymbol{B}_b \in \mathbb{R}^{k\times d_b}$, and mouse random intercept $\boldsymbol{u}_{bm} \in \mathbb{R}^{d_b}$, define

$$\boldsymbol{\xi}_{bmc} = X_{bmc}\boldsymbol{B}_b + \boldsymbol{u}'_{bm} \in \mathbb{R}^{1\times d_b}, \qquad \boldsymbol{\mu}_{bmc} = \mathrm{softmax}\{(\boldsymbol{\xi}_{bmc}, 0)\} \in \Delta^{S_b - 1}.$$

The final path is the reference category. The model is

$$\widetilde{\boldsymbol{x}}_{bmc} \mid \boldsymbol{u}_{bm} \sim \mathrm{Multinomial}(\widetilde{n}_{bmc}, \boldsymbol{\mu}_{bmc}), \qquad \boldsymbol{u}_{bm} \sim N(\boldsymbol{0}_{d_b}, \sigma_b^2 I_{d_b}),$$

where $\sigma_b^2 \in \mathbb{R}_+$ is re-estimated under both null and alternative. An isotropic random-effect covariance is used because some blocks have few mice; an unstructured $d_b \times d_b$ covariance would introduce $d_b(d_b+1)/2$ variance parameters per block.

Let $\ell_{bm}(\boldsymbol{u}; \boldsymbol{B}_b, \sigma_b) \in \mathbb{R}$ be mouse m's conditional multinomial log likelihood plus its Gaussian log density, and let

$$\widehat{\boldsymbol{u}}_{bm} = \arg\max_{\boldsymbol{u}\in\mathbb{R}^{d_b}} \ell_{bm}(\boldsymbol{u}; \boldsymbol{B}_b, \sigma_b), \qquad H_{bm} = -\left.\frac{\partial^2 \ell_{bm}}{\partial \boldsymbol{u}\,\partial \boldsymbol{u}'}\right|_{\widehat{\boldsymbol{u}}_{bm}} \in \mathbb{S}^{d_b}_{+}.$$

The Laplace log marginal likelihood, up to constants common to nested models, is

$$\ell_b^{\text{Lap}}(\boldsymbol{B}_b, \sigma_b) = \sum_{m=1}^{M} \left[\ell_{bm}(\widehat{\boldsymbol{u}}_{bm}; \boldsymbol{B}_b, \sigma_b) + \frac{d_b}{2}\log(2\pi) - \frac{1}{2}\log|H_{bm}|\right] \in \mathbb{R}.$$

The implementation differentiates this approximation through 30 damped Newton updates for every $\widehat{\boldsymbol{u}}_{bm}$. The null contains cell-type intercepts; the alternative appends r_b condition columns. The likelihood-ratio statistic has nominal degrees of freedom $r_b d_b$.

GLMM with EC counts as observations

The path-count GLMM above treats estimated effective path counts as its response. A second model instead retains the observed equivalence-class counts. For gene g, let $T_g \in \mathbb{N}$ be its number of supported isoforms, $d_g = T_g - 1$, $I_g \in \mathbb{N}$ the number of retained pseudobulks, $M_g \in \mathbb{N}$ the number of mice, and $P = 2$ the primer count. Primer p has $K_{gp} \in \mathbb{N}$ gene-assigned ECs. Define the observed counts $\boldsymbol{c}_{gip} \in \mathbb{N}_0^{K_{gp}}$, total $n_{gip} = \mathbf{1}'_{K_{gp}} \boldsymbol{c}_{gip} \in \mathbb{N}_0$, and fixed compatibility or conditional-weight matrix $A_{gp} \in \mathbb{R}_+^{K_{gp} \times T_g}$. Unsupported annotated isoforms and ECs not uniquely assigned to g are excluded.

Let $X_g \in \mathbb{R}^{I_g \times q}$ be the fixed-effect design, $B_g \in \mathbb{R}^{q \times d_g}$ its coefficient matrix, and $u_{gm} \in \mathbb{R}^{d_g}$ mouse m's random isoform-logit intercept. With mouse index $m(i) \in \{1, \dots, M_g\}$, the reference-isoform parameterization is

$$\eta_{gi} = \{X_{gi} B_g + u'_{g,m(i)}, 0\} \in \mathbb{R}^{T_g}, \qquad u_{gm} \sim N(0, \sigma_g^2 I_{d_g}).$$

The probability of EC k from primer p is

$$q_{gipk}(\eta_{gi}) = \frac{\sum_{t=1}^{T_g} A_{gpkt} \exp(\eta_{git})}{\sum_{h=1}^{K_{gp}} \sum_{t=1}^{T_g} A_{gpht} \exp(\eta_{git})}, \qquad \boldsymbol{q}_{gip} \in \Delta^{K_{gp}-1}.$$

Thus the two primers share B_g and u_{gm}, but retain distinct EC maps and positional-coverage weights. The conditional model is either

$$\boldsymbol{c}_{gip} \mid u_{g,m(i)} \sim \text{Multinomial}(n_{gip}, \boldsymbol{q}_{gip})$$

or

$$\boldsymbol{c}_{gip} \mid u_{g,m(i)} \sim \text{DirichletMultinomial}(n_{gip}, \kappa_g \boldsymbol{q}_{gip}), \qquad \kappa_g \in \mathbb{R}_+.$$

The latter adds pseudobulk-level EC overdispersion. It is a direct Dirichlet–multinomial over EC categories, not a Dirichlet model over latent isoform allocations.

An alternative overdispersion model retains the multinomial observation but adds an isotropic observation-level logistic-normal effect. For pseudobulk i, let $e_{gi} \in \mathbb{R}^{d_g}$ and replace the linear predictor by

$$\eta_{gi} = \{X_{gi} B_g + u'_{g,m(i)} + e'_{gi}, 0\}, \qquad e_{gi} \sim N(0, \tau_g^2 I_{d_g}), \quad \tau_g \in \mathbb{R}_+.$$

The same e_{gi} enters both primer likelihoods, because it perturbs the underlying isoform composition rather than each primer's EC probabilities independently. Consequently this model allows correlated extra-multinomial variation across the two primer halves. It is a logistic-normal analogue of the Dirichlet–multinomial, not an identical distribution: its dispersion is isotropic in reference-logit coordinates, whereas the Dirichlet–multinomial has one concentration on the EC simplex.

Laplace inference maximizes the joint conditional density over $u_g \in \mathbb{R}^{M_g \times d_g}$. Its negative Hessian satisfies $H_g \in \mathbb{S}_+^{M_g d_g}$, and the marginal approximation adds the correction $-\frac{1}{2} \log |H_g|$. The outer fit optimizes B_g, σ_g, and, for the overdispersed model, κ_g.

For variational inference, use the diagonal Gaussian family

$$Q(u_g) = \prod_{m=1}^{M_g} \prod_{j=1}^{d_g} N(u_{gmj}; \mu_{gmj}, s_{gmj}^2), \quad \mu_g, s_g \in \mathbb{R}^{M_g \times d_g}, \quad s_{gmj} > 0.$$

For the multinomial likelihood, latent EC-to-isoform responsibilities $r_{gipkt} \in [0,1]$, with $\sum_t r_{gipkt} = 1$ on the support of A_{gpkt}, give the optimized lower bound

$$E_Q\left[\log \sum_t A_{gpkt} e^{\eta_{git}}\right] \geq \log \sum_t A_{gpkt} e^{E_Q \eta_{git}}.$$

The normalizer needs an upper bound. For one observation and primer, let $m_{gipt} = E_Q\eta_{git} + \log \sum_k A_{gpkt}$, $v_{gipt} = \mathrm{Var}_Q(\eta_{git})$, and $a_{gip} \in \Delta^{T_g-1}$. The tilted softmax bound of Knowles and Minka is

$$E_Q \log \sum_{t=1}^{T_g} e^{\eta_{git} + \log \sum_k A_{gpkt}} \leq \frac{1}{2} \sum_t a_{gipt}^2 v_{gipt} + \log \sum_t \exp\left\{ m_{gipt} + \frac{1 - 2a_{gipt}}{2} v_{gipt} \right\},$$

where the local coordinate satisfies

$$a_{gip} = \mathrm{softmax}\{m_{gip} + \tfrac{1}{2}(1 - 2a_{gip}) \odot v_{gip}\}.$$

Alternating the analytic responsibilities, this local fixed point, and the global Gaussian parameters maximizes a tractable lower bound on the multinomial evidence. The cited method is non-conjugate variational message passing rather than conjugate CAVI; here "tilted CAVI" refers only to this block-coordinate implementation.

The full-covariance extension factorizes only across mice. If mouse m has $q_{gm} \in \mathbb{N}$ observed pseudobulks, define its complete latent block

$$z_{gm} = (u'_{gm}, e'_{gi_1}, \ldots, e'_{gi_{q_{gm}}})' \in \mathbb{R}^{L_{gm}}, \qquad L_{gm} = (q_{gm} + 1)d_g,$$

where the e components are omitted in the model without observation noise. Its prior covariance is diagonal,

$$P_{gm} = \mathrm{diag}(\sigma_g^2 I_{d_g}, \tau_g^2 I_{q_{gm}d_g}) \in \mathbb{S}_+^{L_{gm}},$$

but its posterior approximation is unrestricted within mouse,

$$Q(z_g) = \prod_{m=1}^{M_g} N(z_{gm}; \mu_{gm}, \Sigma_{gm}), \quad \mu_{gm} \in \mathbb{R}^{L_{gm}}, \quad \Sigma_{gm} \in \mathbb{S}_{++}^{L_{gm}}.$$

Thus full covariance describes posterior dependence among isoform logits and between a shared mouse effect and its pseudobulk residuals; it does not add an unstructured covariance parameter to the generative model.

For a block-specific cell-type test, the EC likelihood still contains all T_g supported isoforms of gene g; the response is not reduced to path counts. Let $J_b \in \mathbb{N}$ be the number of represented cell types, $Z_b \in \mathbb{R}^{I_g \times q_0}$ the one-hot condition design, and $W_b \in \mathbb{R}^{I_g \times (J_b - 1)}$ the treatment-coded cell-type design. Define $D_b \in \mathbb{R}^{T_g \times d_b}$ by assigning row t the corresponding row of Q_b when isoform t takes a represented path through block b, and assigning it $\mathbf{0}'_{d_b}$ when the isoform is outside the block map. Center the columns of D_b, and let $N_b^{\mathrm{full}} \in \mathbb{R}^{T_g \times (T_g - S_b)}$ be an orthonormal basis for the subspace orthogonal to both its columns and $\mathbf{1}_{T_g}$. In reference-isoform coordinates, define

$$H_b = D_{b,1:(T_g-1),:} - \mathbf{1}_{T_g-1} D_{b,T_g,:} \in \mathbb{R}^{d_g \times d_b}, \qquad N_b = N^{\mathrm{full}}_{b,1:(T_g-1),:} - \mathbf{1}_{T_g-1} N^{\mathrm{full}}_{b,T_g,:} \in \mathbb{R}^{d_g \times (T_g - S_b)}.$$

The free-logit predictor is

$$\eta_{gi,-} = Z_{bi} B_{g0} + \sum_{j=1}^{J_b - 1} W_{bij}(N_b \boldsymbol{\alpha}_{bj} + H_b \boldsymbol{\gamma}_{bj}) + u_{g,m(i)} + e_{gi} \in \mathbb{R}^{d_g},$$

where $B_{g0} \in \mathbb{R}^{q_0 \times d_g}$ contains unrestricted condition effects, $\boldsymbol{\alpha}_{bj} \in \mathbb{R}^{T_g - S_b}$ contains cell-type nuisance effects orthogonal to the tested block directions, and $\boldsymbol{\gamma}_{bj} \in \mathbb{R}^{d_b}$ is cell type j's block-path contrast. The e_{gi} term is present only in the logistic-normal sensitivity model. The null estimates every $\boldsymbol{\alpha}_{bj}$ but sets all $\boldsymbol{\gamma}_{bj} = \mathbf{0}_{d_b}$; the alternative has $(J_b - 1)d_b$ additional coefficients. The combined columns of N_b and H_b span all d_g isoform-logit directions, so the alternative permits unrestricted cell-type isoform composition while the null absorbs every direction orthogonal to the tested path contrast. This construction makes the EC GLMM a per-block test while preserving gene-wide ambiguity and nuisance mass.

For row i, let $R_{gi} \in \{0,1\}^{d_g \times L_{g,m(i)}}$ select and sum its mouse and observation effects. Its free-logit covariance is $R_{gi} \Sigma_{g,m(i)} R'_{gi} \in \mathbb{S}_+^{d_g}$; append a zero row and column for the reference isoform to obtain $V_{gi} \in \mathbb{S}_+^{T_g}$. The correlated tilted bound replaces the variance vector in the diagonal expression. For $m \in \mathbb{R}^{T_g}$, $V \in \mathbb{S}_+^{T_g}$, and $a \in \Delta^{T_g - 1}$,

$$E \log \sum_t e^{\eta_t} \le \frac{1}{2} a' V a + \log \sum_t \exp\{m_t + \tfrac{1}{2} V_{tt} - (Va)_t\}, \qquad a = \operatorname{softmax}\{m + \tfrac{1}{2} \operatorname{diag}(V) - Va\}.$$

The Gaussian KL term is analytic:

$$\mathrm{KL}(Q_m \| P_m) = \frac{1}{2} \left[\operatorname{tr}(P_m^{-1} \Sigma_m) + \mu'_m P_m^{-1} \mu_m - L_{gm} + \log|P_m| - \log|\Sigma_m| \right].$$

Each positive-definite Σ_m is represented by an unconstrained lower Cholesky factor with exponentiated diagonal. The full fit warm-starts from a diagonal variational posterior by copying its means and marginal variances and setting new off-diagonal entries to zero.

The direct EC Dirichlet–multinomial contains $\log \Gamma\{\kappa_g q_{gipk}(u) + c_{gipk}\}$, so the tilted softmax bound does not yield coordinate updates for it. Both likelihood families are therefore also fit with reparameterized Monte Carlo VI. If $w(u) = p(c,u)/Q(u)$, the optimized objective is the ELBO

$$\mathcal{L}_1 = E_Q \log w(u)$$

or the variational Rényi bound of order $\alpha \in (0,1)$,

$$\mathcal{L}_\alpha = \frac{1}{1-\alpha} \log E_Q\{w(u)^{1-\alpha}\}.$$

The comparison uses $\alpha = 0.5$, antithetic Gaussian draws, and the same diagonal or full Gaussian family. Conditional on B_g, σ_g, κ_g, both the likelihood and Q factor by mouse. Writing the corresponding ratio as $w_m(u_{gm})$, the implementation evaluates the equivalent expression

$$\mathcal{L}_\alpha = \sum_{m=1}^{M_g} \frac{1}{1-\alpha} \log E_{Q_m}\{w_m(u_{gm})^{1-\alpha}\}.$$

This avoids the exponential importance-weight degeneracy caused by forming one product weight over all mice. Rényi VI is not called CAVI because its importance-weighted objective has no coordinate update under this model. For the full posterior with observation-level noise, directly optimizing a fixed set of Monte Carlo draws is ill-conditioned: the Cholesky factor can overfit sample correlations when L_{gm} is not small relative to the draw count. This variant is therefore fit with the analytic tilted objective; fresh antithetic draws evaluate its exact-likelihood ELBO and Rényi bound.

The block statistic is twice the optimized alternative-minus-null variational objective. Its reference distribution is empirical rather than χ^2, because it compares optimized evidence bounds and can inherit approximation error. A synchronized label permutation is invalid here because the block null permits real cell-type nuisance effects in orthogonal isoform directions. Instead, a parametric bootstrap simulates mouse effects and optional observation effects from the fitted block null, generates paired-primer EC counts with the observed row totals, and refits both nested models. Each observed statistic is compared with a leave-block-out pool of bootstrap statistics matched by likelihood family, tested dimension $(J_b - 1)d_b$, and gene-depth quartile. Tested dimensions represented by fewer than 20 blocks are combined across depth into a rare-dimension stratum within the same likelihood. With null pool $\mathcal{N}_{-b}$,

$$p_b = \frac{1 + \sum_{z \in \mathcal{N}_{-b}} \mathbf{1}\{z \geq T_b\}}{1 + |\mathcal{N}_{-b}|}.$$

Every bootstrap statistic is calibrated by the same leave-block-out rule. Benjamini–Hochberg values are reported for a likelihood only if these calibrated null p-values reject between 2.5% and 7.5% at level 0.05.

Liang–Zeger multinomial GEE

GEE specifies only the marginal mean and first two moments. It can therefore use the fractional effective counts directly. Define $\boldsymbol{p}_{bmc} = \boldsymbol{x}_{bmc}/n^{\text{eff}}_{bmc} \in \Delta^{S_b - 1}$, and let $\boldsymbol{p}_{bmc,-} \in \mathbb{R}^{d_b}$ omit the reference path. Its marginal mean uses the same reference-logit regression,

$$\boldsymbol{\mu}_{bmc} = \operatorname{softmax}\{(X_{bmc}\boldsymbol{B}_b, 0)\}, \qquad G_{bmc} = \operatorname{diag}(\boldsymbol{\mu}_{bmc,-}) - \boldsymbol{\mu}_{bmc,-}\boldsymbol{\mu}'_{bmc,-} \in \mathbb{S}^{d_b}_{+}.$$

The working covariance of one proportion vector is $A_{bmc} = G_{bmc}/n^{\text{eff}}_{bmc} \in \mathbb{S}^{d_b}_{+}$. Let $L_{bmc} \in \mathbb{R}^{d_b \times d_b}$ be its symmetric square root, and let

$$D_{bmc} = \frac{\partial \boldsymbol{\mu}_{bmc,-}}{\partial \operatorname{vec}(\boldsymbol{B}_b)'} = [X_{bmc,1}G_{bmc} \ \cdots \ X_{bmc,k}G_{bmc}] \in \mathbb{R}^{d_b \times kd_b}.$$

For mouse m, stack its $q_m \in \mathbb{N}$ observed cell types into $\boldsymbol{p}_{bm,-} \in \mathbb{R}^{q_m d_b}$, mean $\boldsymbol{\mu}_{bm,-} \in \mathbb{R}^{q_m d_b}$, and derivative $D_{bm} \in \mathbb{R}^{q_m d_b \times kd_b}$. With $L_{bm} = \operatorname{blockdiag}_c(L_{bmc})$ and exchangeable $R_m(\alpha_b) \in \mathbb{R}^{q_m \times q_m}$, the Liang–Zeger working covariance is

$$W_{bm} = \phi_b L_{bm}\{R_m(\alpha_b) \otimes I_{d_b}\}L'_{bm} \in \mathbb{S}^{q_m d_b}_{+}.$$

The scale $\phi_b \in \mathbb{R}_+$ and working correlation $\alpha_b \in (-1/(q_{\max} - 1), 1)$ are updated from whitened Pearson residuals. Coefficients solve

$$\sum_{m=1}^{M} D'_{bm}W^{-1}_{bm}(\boldsymbol{p}_{bm,-} - \boldsymbol{\mu}_{bm,-}) = \mathbf{0}_{kd_b}.$$

If $\boldsymbol{U}_{bm} = D'_{bm}W^{-1}_{bm}(\boldsymbol{p}_{bm,-} - \boldsymbol{\mu}_{bm,-}) \in \mathbb{R}^{kd_b}$ and $\mathcal{H}_b = \sum_m D'_{bm}W^{-1}_{bm}D_{bm} \in \mathbb{S}^{kd_b}_{+}$, the canonical cluster-robust covariance is

$$\widehat{\operatorname{Var}}\{\operatorname{vec}(\widehat{\boldsymbol{B}}_b)\} = \mathcal{H}^{+}_b \left(\sum_{m=1}^{M} \boldsymbol{U}_{bm}\boldsymbol{U}'_{bm}\right) \mathcal{H}^{+}_b \in \mathbb{S}^{kd_b}_{+}.$$

The condition coefficients are tested by a robust Wald statistic. No small-sample sandwich correction is imposed; synchronized mouse-level permutations assess whether the canonical large-M reference is adequate.

Matched power benchmark

Power is evaluated only for partitions represented by exactly two conditions. For replicate r, condition labels are first permuted among mice. Let $z_{bmc}^{(r)} \in \{0,1\}$ be the resulting non-reference condition indicator and let $\boldsymbol{v}_b \in \mathbb{R}^{d_b}$ be a reproducible random unit vector. For ILR effect $\gamma \in \{0.25, 0.5\}$, define

$$\boldsymbol{y}_{bmc}^{(r,\gamma)} = \boldsymbol{y}_{bmc} + \gamma z_{bmc}^{(r)} \boldsymbol{v}_b, \qquad \boldsymbol{\psi}_{bmc}^{(r,\gamma)} = \operatorname{softmax}\{Q_b \boldsymbol{y}_{bmc}^{(r,\gamma)}\}.$$

The shifted effective counts are $n_{bmc}^{\text{eff}} \boldsymbol{\psi}_{bmc}^{(r,\gamma)}$. Thus every method sees the same imposed ILR effect, observed effective depth, cell-type pattern, and mouse clustering. For each simulated alternative, null-calibrated power uses the finite-permutation rank of its statistic among synchronized null statistics for the same block and method. This event-matched calibration preserves dimension, depth, and cluster structure.

Algorithms

All algorithms operate independently over canonical blocks unless a joint gene-level combination is stated. "Fit" below means bounded L-BFGS with analytic gradients and a fourfold iteration-limit retry after a warm start from the first endpoint.

Algorithm 1 Construct splice blocks and path maps

Require: K_g annotated spliced isoforms with exon chains; retained set $\mathcal{I}_g$, $|\mathcal{I}_g| = T_g$
Ensure: Blocks b, maps $C_b \in \{0,1\}^{S_b \times T_g}$, and bases $Q_b \in \mathbb{R}^{S_b \times d_b}$

Split the gene at every annotated exon boundary into atomic intervals.
Merge adjacent intervals exonic in every annotated isoform into constitutive anchors.
Form internal blocks between consecutive anchors and terminal blocks using pseudo-anchors.
for all candidate blocks b **do**
    Record each annotated isoform's ordered intervening exon chain.
    Assign identical chains to one path and restrict to $\mathcal{I}_b^{\text{sp}} \subseteq \mathcal{I}_g$.
    Remap represented path labels to $\{1, \ldots, S_b\}$.
    if $S_b < 2$ **then**
        discard b
    else
        Set $(C_b)_{s_b(t),t} = 1$ for $t \in \mathcal{I}_b^{\text{sp}}$; leave nuisance columns zero.
        Construct the Helmert basis Q_b.
    end if
end for
Canonicalize the unordered isoform partition and retain one representative of equivalent blocks.

Algorithm 2 Fit one pseudobulk's local path composition

Require: Counts $\boldsymbol{c}_{bip} \in \mathbb{N}_0^{E_{bp}}$, designs $A_{bp} \in \mathbb{R}_+^{E_{bp} \times T_g}$, baseline $\boldsymbol{\theta}_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$, C_b, and Q_b

Ensure: $\widehat{\boldsymbol{\theta}}_{bi}$, $\widehat{\boldsymbol{\psi}}_{bi}$, $\widehat{\boldsymbol{y}}_{bi}$, and V_{bi}^{cond}

Initialize $\boldsymbol{\delta}_{bi} \leftarrow \mathbf{0}_{d_b}$.
if $S_b = 2$ **then**
    Evaluate the fixed log-ratio grid and retain its best point.
end if
function OBJECTIVE($\boldsymbol{\delta}$)
    Form $\boldsymbol{\theta}_{bi}(\boldsymbol{\delta})$, preserving spliced mass and within-path isoform mixtures.
    return the negative sum of the $P = 2$ multinomial log likelihoods and its analytic gradient.
end function
Minimize OBJECTIVE($\boldsymbol{\delta}$) by bounded L-BFGS on $[-20, 20]^{d_b}$.
Set $\widehat{\boldsymbol{\theta}}_{bi} \leftarrow \boldsymbol{\theta}_{bi}(\widehat{\boldsymbol{\delta}}_{bi})$.
Set $\widehat{\boldsymbol{\psi}}_{bi} \leftarrow C_b \widehat{\boldsymbol{\theta}}_{bi} / (\mathbf{1}'_{S_b} C_b \widehat{\boldsymbol{\theta}}_{bi})$.
Set $\widehat{\boldsymbol{y}}_{bi} \leftarrow Q'_b \log \widehat{\boldsymbol{\psi}}_{bi}$.
Compute $I_{\delta,bi}$ by summing primer expected information.
Apply the eigenvalue/null-space check to $(I_{\delta,bi}, I_{d_b})$; if identifiable, set $V_{bi}^{\text{cond}} = I_{\delta,bi}^{-1}$, otherwise set it to infinity.

Algorithm 3 Compute full-isoform nuisance (profile) covariance

Require: $\widehat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{T_g}$, A_{bp}, N_{bip}, C_b, Q_b, relative threshold $r_{\text{eig}} = 10^{-8}$, and null tolerance $a_{\text{null}} = 10^{-9}$

Ensure: $V_{bi}^{\text{prof}} \in \mathbb{S}_+^{d_b}$ and an identifiability flag

for $p = 1, \ldots, P$ **do**
    Compute $\widehat{\boldsymbol{q}}_{bip}$, $\widehat{Z}_{bip}$, and $I_{\eta,bip} \in \mathbb{S}_+^{T_g}$.
end for
Set $I_{\eta,bi} \leftarrow \sum_p I_{\eta,bip}$.
Compute path masses a_{bis}, spliced mass ω_{bi}^{sp}, and $R_{bi} = \partial \log \boldsymbol{\psi}_{bi} / \partial \boldsymbol{\eta}'_{bi}$.
Set $J_{bi} \leftarrow Q'_b R_{bi}$.
Eigendecompose $I_{\eta,bi} = U_{bi} \operatorname{diag}(\boldsymbol{\lambda}_{bi}) U'_{bi}$.
Put t in $\mathcal{P}_{bi}$ when $\lambda_{bit} > r_{\text{eig}} \max_u |\lambda_{biu}|$, and put all other indices in $\mathcal{N}_{bi}$.
if $\|J_{bi} U_{bi,0}\|_F > a_{\text{null}} \max\{\|J_{bi}\|_F, 1\}$ **then**
    Set $V_{bi}^{\text{prof}} \leftarrow \infty_{d_b \times d_b}$ and mark the response unidentified.
else
    Set $V_{bi}^{\text{prof}} \leftarrow J_{bi} U_{bi,+} \operatorname{diag}(\boldsymbol{\lambda}_{bi,\mathcal{P}}^{-1}) U'_{bi,+} J'_{bi}$.
    Mark the response identifiable.
end if

Algorithm 4 Fit a cell-resolved pseudobulk path composition

Require: Cell EC vectors $\boldsymbol{c}_{bihp}$, baselines $\boldsymbol{\theta}^{(0)}_{bih}$, weights w_{bih}, and A_{bp}, C_b, Q_b
Ensure: One pseudobulk response $\widehat{\boldsymbol{y}}_{bi}$ and covariance V_{bi}

1: Allocate one shared $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$.
2: **for all** cells h and primers p **do**
3: Perturb the cell's baseline, normalize its EC probabilities, and add its multinomial log likelihood and gradient.
4: **end for**
5: Optimize the summed objective over $\boldsymbol{\delta}_{bi}$.
6: Compute each fitted $\widehat{\boldsymbol{\psi}}_{bih}$ and their normalized w_{bih}-weighted mean $\widehat{\boldsymbol{\psi}}_{bi}$.
7: Sum cell-by-primer Fisher matrices and propagate their inverse through the weighted composition response to obtain V_{bi}.
8: Set $\widehat{\boldsymbol{y}}_{bi} = Q_b' \log \widehat{\boldsymbol{\psi}}_{bi}$.

Algorithm 5 Gaussian condition or paired cell-type test

Require: Responses $\widehat{\boldsymbol{y}}_{bi}$, covariances V_{bi}, mouse, condition, and cell-type labels; test type
Ensure: One Wald p-value for block b

1: **if** testing condition within a cell type **then**
2: Retain conditions represented by at least three mice.
3: Build full design $X_b \in \mathbb{R}^{n \times k}$ and reduced design $X_{0b} \in \mathbb{R}^{n \times (k - r_b)}$.
4: **else**
5: Build mouse-effect design $X_{0b} \in \{0, 1\}^{n_b \times M_b}$.
6: Append $L_b - 1$ cell-type columns to obtain $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$.
7: **end if**
8: Require full column rank in both designs.
9: Estimate τ_b^2 by REML under the reduced design using $V_{bi} + \tau_b^2 I_{d_b}$.
10: Hold $\widehat{\tau}_b^2$ fixed and fit the full design by GLS.
11: Stack the tested coefficients and compute their Wald statistic W_b.
12: Compare W_b with the applicable chi-squared reference.
13: Calibrate by mouse-level condition permutations or within-mouse cell-type permutations, as appropriate.

Algorithm 6 Paired compositional cell-type test

Require: $\widehat{\boldsymbol{\psi}}_{bi}$, V_{bi}^{cond}, mouse and cell-type labels, and $R = 500$
Ensure: One exact permutation p-value p_b per canonical block
Keep converged, identifiable estimates satisfying the path floor and the three-mouse-per-cell-type requirement.
Convert each $(\widehat{\boldsymbol{\psi}}_{bi}, V_{bi}^{\text{cond}})$ to n_{bi}^{eff} and $\boldsymbol{x}_{bi}$.
Build X_{0b} from mouse effects and X_{1b} by appending cell-type effects.
Fit the null coefficients and $\widehat{\kappa}_{0b}$.
Fit the observed alternative and compute Λ_b.
if using fixed concentration **then**
  Set $\kappa_{1b} \leftarrow \widehat{\kappa}_{0b}$.
else
  Optimize $\widehat{\kappa}_{1b}$ and compare the finite optimum with the exact multinomial endpoint.
end if
for $r = 1, \ldots, R$ **do**
  Shuffle cell-type labels within each mouse to form $X_{1b}^{(r)}$.
  Reuse the null fit, refit the alternative, and store $\Lambda_b^{(r)}$.
end for
Compute the tolerance-aware finite-permutation rank p_b.
Require $n_b \geq 30$, then apply BH across canonical blocks.

Algorithm 7 Liang–Zeger multinomial GEE condition test

Require: Fractional counts $\boldsymbol{x}_{bmc}$, design X_b, mouse labels, and tested condition columns
Ensure: Robust Wald statistic and p-value
Initialize $\boldsymbol{B}_b$ from independent multinomial regression and set $\alpha_b \leftarrow 0$, $\phi_b \leftarrow 1$.
repeat
  Compute $\boldsymbol{\mu}_{bmc}$, G_{bmc}, A_{bmc}, and D_{bmc} for every observation.
  Whiten residuals by $A_{bmc}^{-1/2}$.
  Update ϕ_b from squared Pearson residuals and α_b from within-mouse Pearson cross-products.
  Construct every W_{bm}, quasi-score $\boldsymbol{U}_{bm}$, and sensitivity matrix $\mathcal{H}_b$.
  Update $\operatorname{vec}(\boldsymbol{B}_b) \leftarrow \operatorname{vec}(\boldsymbol{B}_b) + \mathcal{H}_b^{+} \sum_m \boldsymbol{U}_{bm}$.
until the maximum coefficient update is at most 10^{-8}
Form the cluster sandwich covariance $\mathcal{H}_b^{+}(\sum_m \boldsymbol{U}_{bm}\boldsymbol{U}_{bm}')\mathcal{H}_b^{+}$.
Compute the robust Wald test for all condition coefficients.
Permute condition once per mouse and repeat the complete GEE fit.

Algorithm 8 Laplace multinomial GLMM condition test

Require: Effective counts, nested designs, mouse labels, and random-effect dimension d_b
Ensure: Likelihood-ratio statistic and p-value

```
Integerize each effective count vector by largest remainder.
for all null and alternative models do
    Initialize fixed coefficients from multinomial regression and initialize σ_b.
    repeat
        for all mice m do
            Find û_bm with 30 damped Newton updates.
            Compute random-effect Hessian H_bm and its log determinant.
        end for
        Sum the Laplace log marginal likelihood and differentiate through the Newton updates.
        Update B_b and log σ_b by bounded L-BFGS.
    until the outer gradient tolerance or iteration limit is reached
end for
Compute twice the null-to-alternative Laplace log-likelihood gain.
Compare with χ²_{r_b d_b}.
For permutations, reuse the fitted null and refit each permuted alternative.
```

Algorithm 9 Inference comparison for the paired EC-count GLMM

Require: Paired counts $\boldsymbol{c}_{gip}$, maps A_{gp}, nested fixed-effect designs, mouse indices, likelihood family, and Rényi order $\alpha = 0.5$
Ensure: Null and alternative evidence approximations and coefficient estimates

```
for all null and alternative designs do
    Fit the conditional posterior mode of u_g ∈ ℝ^{M_g × d_g} by damped Newton updates.
    Optimize the Laplace marginal likelihood over B_g, log σ_g, and optional log κ_g.
    if the likelihood is multinomial then
        repeat
            Update every latent EC-to-isoform responsibility from A_gpkt exp(E_Q η_git).
            Iterate every tilted coordinate a_gip to its softmax fixed point.
            Update B_g, μ_g, log s_g, log σ_g against the bounded ELBO; damp the global update
            when the bound decreases.
        until the ELBO gradient tolerance or iteration limit is reached
    end if
    Draw fixed antithetic standard-normal samples and reparameterize u_g = μ_g + s_g ⊙ ϵ.
    Optimize the exact-likelihood Monte Carlo ELBO ℒ_1.
    Warm-start from the ELBO fit and optimize ℒ_0.5; report its effective importance-sample
    size.
end for
Compare each method's null-to-alternative gain on its own objective; do not compare raw
objective values across approximation families.
```

Algorithm 10 Block-specific paired EC-count GLMM cell-type test

Require: Block b, paired gene EC counts $\boldsymbol{c}_{gip}$, maps A_{gp}, condition design Z_b, cell-type design W_b, mouse indices, and R parametric-bootstrap replicates

Ensure: Empirically calibrated block p-value

Construct the Helmert basis Q_b and isoform-to-path contrast matrix H_b; retain unmapped isoforms as likelihood nuisances.

Form the null fixed-effect tensor from unrestricted condition-by- isoform coefficients $Z_b \otimes I_{d_g}$ and cell-type nuisance columns $W_{bij}(N_b)_{:s}$.

Fit the null full-covariance posterior and variance parameters.

Append the $(J_b - 1)d_b$ columns $W_{bij}(H_b)_{:s}$, warm-start all null parameters, and fit the observed alternative.

Set $T_b \leftarrow 2(\mathcal{L}_{1b} - \mathcal{L}_{0b})$, using the same variational objective for both nested fits.

for $r = 1, \ldots, R$ **do**

  Draw mouse effects and optional observation effects from the fitted null; compute primer-specific EC probabilities.

  Generate multinomial or Dirichlet–multinomial EC counts while preserving every observed primer-row total.

  Refit the null, warm-starting from the observed null, then append the block-effect columns and fit the nested alternative.

end for

Pool converged bootstrap statistics by likelihood, tested dimension, and depth stratum, excluding block b's replicates.

Compute the finite-pool rank p-value and assess the same rank calculation on every null replicate before applying BH.

Algorithm 11 Matched semi-synthetic power benchmark

Require: Two-condition partitions, effect sizes γ, and replicate count R_{power}

Ensure: Asymptotic and null-calibrated power for each method

for all eligible blocks b **do**

  Draw and fix a unit ILR direction $\boldsymbol{v}_b$.

  for $r = 1, \ldots, R_{\text{power}}$ **do**

    Permute condition labels among mice.

    for all nonzero effect sizes γ **do**

      Set $\boldsymbol{y}_{bmc}^{(r,\gamma)} = \boldsymbol{y}_{bmc} + \gamma z_{bmc}^{(r)} \boldsymbol{v}_b$.

      Map shifted ILRs to compositions and effective path counts.

      Fit GEE, GLMM, clustered GLS, and Dirichlet–multinomial tests.

    end for

  end for

end for

Rank each alternative statistic against null statistics for its block and method, using the finite-permutation correction.

Report asymptotic and event-matched permutation rejection at 0.05.

Algorithm 12 Pairwise and dominant-path sensitivity tests

Require: Path estimates and conditional covariances from Algorithm 2
Ensure: Calibrated sensitivity tables

```
for all cell-type pairs do
    Retain M mice containing both types.
    Use X_0 ∈ ℝ^{2M×M} for mouse effects and X_1 ∈ ℝ^{2M×(M+1)} for mouse plus cell type.
    Permute the two labels within mouse and correct across all block-by-pair hypotheses.
end for
for all blocks b do
    Rank paths by ∑_i N_bi^gene ψ_bis and select S_b, the two largest, without labels.
    Construct selector E_b, two-path basis Q_b^(2), and R_b = (Q_b^(2))′ E_b′ Q_b.
    Set V_bi^(2) = R_b V_bi R_b′ and renormalize the two retained proportions.
    Run Algorithm 6 and record path-floor sensitivity sweeps.
end for
```

Algorithm 13 Joint condition tests

Require: Calibrated block-by-cell-type p-values or ILR responses
Ensure: Partition- or gene-level joint p-values

```
For ACAT, combine valid cell-type p-values within a canonical partition; optionally combine
  partition p-values within a gene.
As a negative control, fit one shared condition effect while treating cell-type rows as indepen-
  dent.
For clustered GLS, construct diagonal covariance blocks V_bmc + (τ²_mouse + τ²_resid) I_{d_b} and same-
  mouse off-diagonal blocks τ²_mouse I_{d_b}.
Estimate τ²_mouse and τ²_resid by REML under the null.
Fit the shared condition coefficient by GLS.
for all mouse-level condition permutations do
    Apply one permuted condition to all cell types from that mouse and refit the model.
end for
Apply BH to the canonical-partition family; keep negative-control and calibrated results sepa-
  rate.
```

Implementation Details

Splice-block construction

GTF exons are represented by zero-based, half-open intervals. Exon boundaries define atomic intervals; intervals exonic in every annotated spliced isoform are merged into constitutive anchors. Consecutive anchors define a block, and an isoform path is its ordered exon-interval chain between them. Terminal pseudo-anchors include alternative first and last regions. A gene without a shared constitutive region receives one whole-gene block.

Blocks with one path are discarded. Simple two-path blocks include skipped exons, mutually exclusive exons, alternative donors and acceptors, and retained introns. The implementation also supports nested and multi-path blocks. Precursor isoforms remain nuisance isoforms in the EC model but are not assigned to spliced paths.

Different genomic blocks can induce exactly the same partition of annotated isoforms, for example when several exons always occur together. Such blocks have identical statistical responses and must not count as repeated evidence. A canonical partition is the unordered collection of isoform sets assigned to paths. Equivalent blocks are collapsed before FDR or joint testing; all member interval identifiers remain in the output.

Pseudobulks and local fitting

Cell isoform proportions from the hierarchical decoder are weighted by cell UMI totals and aggregated by mouse, condition, and cell type. Pseudobulks must contain at least 20 cells and 10^5 total UMIs. A block is fit when its gene has at least 25 gene-assigned UMIs in at least two qualifying pseudobulks. Only ECs whose compatible isoforms map uniquely to that gene enter the local likelihood.

Bounded L-BFGS optimizes the paired multinomial likelihood, with $\|\boldsymbol{\delta}_{bi}\|_\infty \leq 20$ to prevent floating-point underflow, where $\|\cdot\|_\infty$ is the maximum absolute component. Two-path fits are initialized from a log-ratio grid because the paired-primer normalizers can make the objective non-concave.

For the cell-resolved variant, sparse cell-by-EC matrices are retained and one local isoform matrix is decoded at a time. The optimizer dimension is still only the number of paths minus one. Cell-specific Fisher contributions are accumulated analytically, so the method does not fit one free splicing parameter per cell.

Identifiability and reliability

Finite Fisher covariance is not sufficient near a simplex boundary. Parametric bootstrap identified two reliability requirements:

$$\lambda_{\max}(V_{bi}) \leq 1/8 \quad \text{and} \quad \min_s \widehat{\psi}_{bis} \geq 0.15.$$

Thus every tested log-ratio direction has effective information of at least eight and every fitted path has at least 15% usage. All estimates are written, together with convergence, identifiability, and reliability flags; only reliable estimates enter GLS. Here $\lambda_{\max}(V_{bi}) \in \mathbb{R}_+$ is the largest eigenvalue of the $d_b \times d_b$ covariance, and $\widehat{\psi}_{bis} \in [0,1]$ is path s's fitted proportion.

Results

Condition-test scope

The primary analysis used the 25-gene-UMI cutoff and blocks with at most eight represented paths. It fit 10,100 blocks and 359,713 pseudobulk-by-block path estimates; 359,711 fits converged. After reliability filtering, there were 829 conditional and 506 profile tests.

These 1,335 rows represent 829 distinct block-by-cell-type hypotheses, covering 527 distinct splice blocks in eight cell types. The profile analysis is available for a subset of the conditional hypotheses. The possible conditions were 5xFAD, 5xFIRE, 5xFIRE PBS, 5xFIRE transplant, FIRE, and WT. For the 829 conditional hypotheses, 275 compared two conditions, 155 compared three, 153 compared four, 130 compared five, and 116 compared all six. The included set varies because every condition in a test must have at least three independent mouse pseudobulks for that block and cell type.

Condition-test results

No block was significant at 5% FDR in either covariance analysis. At the unadjusted $p < 0.05$ level:

- the conditional analysis identified 24 block-by-cell-type hypotheses, covering 21 distinct blocks;
- the profile analysis identified 20 block-by-cell-type hypotheses, covering 17 distinct blocks;
- 20 hypotheses were nominally significant in both analyses.

The union is 24 nominal block-by-cell-type hypotheses covering 21 blocks, but the corrected number of significant hypotheses and blocks is zero. Thus there is no FDR-significant evidence for condition-dependent splicing within cell type. The two covariance variants should not be added as if they were separate discoveries.

Formal mouse-clustered models

The shared-condition analysis was repeated with the multinomial-logit GLMM and Liang–Zeger GEE defined above. Clustered Gaussian ILR GLS and the unclustered Dirichlet–multinomial were refit to the same canonical partitions as comparators. Each block used 50 synchronized mouse-level null permutations. For 88 two-condition blocks, 50 semi-synthetic replicates were generated at each ILR effect size; 4,369 of 4,400 replicate designs had full rank. "Perm. nominal" below is the number of observed partitions with an event-matched permutation p-value at most 0.05. Table 1 summarizes the complete analysis.

Table 1: Calibration and power comparison. Null rejection uses asymptotic p-values on permuted labels; power uses event-matched permutation ranks.

Method	Tests	Conv.	Null $< .05$	Perm. nominal	Power .25	Power .50
Clustered GLS	359	359	0.025	17	0.171	0.322
Dirichlet–multinomial	359	359	0.138	18	0.188	0.364
Liang–Zeger GEE	359	355	0.415	8	0.139	0.268
Laplace GLMM	359	359	0.108	19	0.199	0.390

The canonical GEE sandwich reference is unusable at the available mouse counts: 41.5% of permuted-null p-values were below 0.05, and its apparent 116 asymptotic FDR discoveries are artifacts of this inflation. The GLMM reference was closer but still anti-conservative at 10.8%, while clustered GLS was conservative at 2.5%. Event-matched permutation calibration is therefore required for all comparisons. On that scale, the GLMM had the highest power, improving over clustered GLS by 2.8 and 6.9 percentage points at ILR effects 0.25 and 0.5. GEE was less powerful than clustered GLS after calibration. No method produced a permutation-calibrated 5% FDR discovery in the observed condition analysis.

EC-count GLMM inference benchmark

The EC-count methods were first compared in 96 independent simulations per likelihood and effect size. Each replicate had 12 mice, two observations per mouse, two isoforms, two distinct three-EC primer maps, 150 molecules per primer and observation, and mouse random-effect standard deviation 0.4. Dirichlet–multinomial simulations used concentration 30. The null threshold for each method was its own 95th percentile evidence gain at effect zero; power was evaluated at an isoform logit effect of 0.5. This calibration is required because Laplace likelihoods, tilted ELBOs, and Rényi bounds are not numerically interchangeable objectives.

Table 2: Ambiguous-EC simulation comparison. RMSE is for the condition logit coefficient. Time is median seconds per nested null/alternative fit at effect 0.5. ESS is the median alternative mouse-level importance ESS out of 128.

Likelihood	Approximation	Null rejection	Power	RMSE	Time	ESS
Multinomial	Laplace	0.052	0.542	0.223	9.87	–
Multinomial	Tilted ELBO	0.052	0.542	0.223	1.87	–
Multinomial	Rényi $\alpha = 0.5$	0.052	0.542	0.223	1.58	127.67
Dirichlet–multinomial	Laplace	0.052	0.417	0.236	21.41	–
Dirichlet–multinomial	Monte Carlo ELBO	0.052	0.417	0.236	3.06	127.51
Dirichlet–multinomial	Rényi $\alpha = 0.5$	0.052	0.417	0.236	3.35	127.66

Within each likelihood family, all three approximations had the same empirically calibrated power to the reported precision, and their coefficient bias differed by less than 2×10^{-4}. The tilted bound's null cutoff was 4.70, compared with 4.60 for multinomial Laplace and Rényi VI, but this small scale difference did not change rejection. Variational fitting was five- to seven-fold faster than Laplace. A single global importance weight over all mice had ESS 1/128 and collapsed the fitted random-effect variance; the mouse-factorized Rényi objective corrected this failure and produced ESS above 127.5/128 for both likelihoods.

A second simulation used three isoforms and otherwise retained the same 96 null and 96 effect replicates. Full-covariance and diagonal VI were nearly indistinguishable. Tilted evidence gains had Spearman correlation 0.9999 and median absolute difference 0.073 between posterior families; Rényi gains had correlation 0.99996 and median absolute difference 0.044. Every VI variant had calibrated power 0.438 and condition-coefficient RMSE 0.237–0.238. The full-covariance tilted fit took a median 2.43 seconds per nested effect fit versus 2.00 seconds for diagonal tilted VI. Thus, in a controlled multi-isoform setting, omitted posterior covariance was not a material source of error.

The same conclusion held for the direct EC Dirichlet–multinomial. In 96 three-isoform null and effect simulations, diagonal and full ELBO gains had correlation 0.9997 and median absolute difference 0.150; diagonal and full Rényi gains had correlation 0.9999 and median difference 0.092. Both posterior families had calibrated power 0.219. In the 65-gene real-data analysis, diagonal and full ELBO gains had correlations 0.998 for condition and 0.9998 for cell type, with median absolute differences 0.0035 and 0.0050. Fixed-draw Rényi optimization was again less stable than ELBO optimization.

The genome-wide paired-primer comparison retained 65 genes with mean depth at least 100 EC UMIs per pseudobulk; this was 65 of 66 depth-eligible genes, with one 3,905-EC outlier excluded by the 128-EC computational cap. All retained genes had at most six EC-supported isoforms. The authoritative run initialized each Gaussian variational posterior from the corresponding Laplace means and covariance diagonal, then initialized Rényi VI from the ELBO fit. Across the 65 condition models, convergence was 98.5% for both Laplace likelihoods, 95.4% for both ELBO variants, 92.3% for multinomial Rényi VI, and 98.5% for Dirichlet–multinomial Rényi VI. Cell-type convergence ranged from 93.8% to 100%. There were no fit exceptions.

Approximation agreement depended on isoform dimension. Among the 38 two-isoform genes, multinomial Laplace and Rényi evidence gains had Spearman correlation 0.974 for condition and 0.9996 for cell type; Laplace and tilted ELBO correlations were 0.969 and 0.9995. All 16 two-isoform condition genes above the method-specific nominal threshold under multinomial Laplace were also above it under Rényi VI. For the 20 three-isoform genes, the analogous gain correlations fell to 0.49 and 0.59. The mean-field posterior has only a diagonal covariance across isoform random effects, whereas Laplace retains their within-mouse posterior covariance. The dimension-dependent divergence therefore motivated a full-covariance extension, but by itself did not distinguish covariance error from objective or optimizer differences. One-dimensional simulation thresholds were not applied to multi-isoform or cell-type omnibus tests.

The tilted surrogate must also be separated from objective evaluation. For two-isoform condition models, median nested gains were 5.54 for the tilted training bound, 4.95 for an exact-likelihood Monte Carlo ELBO evaluated at the same posterior, 4.93 for its Rényi evaluation, 4.93 for Laplace, and 4.95 for separately optimized Rényi VI. For cell type, the corresponding medians were 89.06, 74.38, 74.36, 72.12, and 74.36. Exact ELBO and Rényi gains on the same posterior had Spearman correlations 0.998 and 1.000 for condition and cell type. The tilted bound becomes looser under the larger alternative and therefore overstates nested evidence gains; it is retained as a fitting surrogate, not used as a likelihood-ratio statistic.

Full-covariance results did not support posterior covariance as the cause of the earlier multi-isoform discrepancy. On the real condition analysis, fresh-sample ELBO gains from diagonal and full tilted posteriors had Spearman correlation 0.991 across 65 genes and median absolute difference 0.002. Among the 20 three-isoform genes, the correlation was 0.938 and median difference was 0.024. The corresponding cell-type values were 1.000 overall and within the three-isoform subset, with median three-isoform difference 0.023. These results, together with the three-isoform simulation, point instead to differences in the Laplace objective or its outer optimization. Separately optimizing a fixed-draw full-covariance Rényi objective was less stable than fitting the tilted objective and evaluating both exact objectives with fresh draws; the latter is the reported comparison.

The observation-level logistic-normal model was validated with 96 null and 96 effect simulations using three isoforms, true residual standard deviation 0.6, and the same ambiguous paired-primer maps. Tilted full-covariance VI estimated median residual standard deviation 0.559 under the effect and had calibrated power 0.354. Dirichlet–multinomial Laplace had power 0.365, while the misspecified ordinary multinomial had power 0.313. Logistic-normal Laplace had power 0.146 despite estimating residual scale 0.472, indicating that its high-dimensional marginal approximation was not competitive with the tilted fit.

In the real data, tilted logistic-normal fits converged under both nested models for 80.0% of condition genes and 86.2% of cell-type genes. The alternative residual standard deviation had median 0.040 for condition and 0.029 for cell type; the respective interquartile ranges were 0.007–0.239 and 0.006–0.234. Relative to the multinomial full posterior, fresh ELBO condition gains changed by a median absolute 0.335 and retained rank correlation 0.809. Cell-type gains changed by 10.68 and retained correlation 0.959, consistent with extra-multinomial variation having more impact on the larger cell-type omnibus statistic. Because residual-scale estimates often approached zero and convergence was weaker, this model is retained as an overdispersion sensitivity analysis rather than replacing the primary test.

The block-specific EC analysis evaluated the 406 coverage-eligible canonical partitions from the primary cell-type analysis. Seven had fewer than two paths represented in the EC design, leaving 399 identifiable tests. Five parametric-bootstrap replicates per block supplied approximately 2,000 null statistics per likelihood; only converged nested refits entered the pooled calibration. Table 3 reports the conditional block tests after including orthogonal cell-type isoform effects in every null.

Table 3: Parametric-bootstrap-calibrated cell-type tests using gene EC counts. Primary overlap counts discoveries also found by the 101-event exact path-count analysis.

Likelihood	Tests	Converged	Null < .05	Nominal	FDR	Primary overlap
Multinomial	399	395	0.0480	111	73	29
Logistic-normal multinomial	399	362	0.0486	104	70	29
Dirichlet–multinomial	399	389	0.0494	70	23	7

The multinomial and logistic-normal results were similar: 66 discoveries were shared, and they represented 66 and 63 genes, respectively. The direct Dirichlet–multinomial was more conservative, with 23 discoveries in 21 genes; 16 were shared by all three EC likelihoods. All three empirical null rates were within 0.2 percentage points of 0.05, so BH correction was enabled for each method. Agreement with the path-count analysis was limited rather than complete. This is expected to reflect both retained EC ambiguity and the different null: the EC test conditions on arbitrary cell-type isoform effects orthogonal to the local block direction.

Cell-type effects within mouse

The initial Gaussian analysis tested 369 conditional and 218 profile partitions. It found 22 and five FDR-significant partitions, respectively, with all profile discoveries contained in the conditional set. After canonicalizing annotation intervals that encode the same isoform-to-path partition, these 22 intervals represented 14 distinct partitions.

The compositional analysis used converged, conditionally identifiable estimates and required every modeled path proportion to be at least 5%. This produced 646 nonredundant tests. A minimum of 30 pseudobulk observations per partition, chosen as a label-effect-independent coverage requirement, retained 406 hypotheses for FDR correction. Among these, 149 had exact permutation $p < 0.05$, and **101 splice-path partitions in 95 genes passed 5% FDR**. Thirteen of the 14 nonredundant Gaussian discoveries were recovered; the remaining partition had exact $p = 0.032$ but FDR 0.098. Significant compositional results had median maximum cell-type logit effect 0.84 and median effective path count 21.7.

The observation-count threshold is not a narrow boundary effect: thresholds from 29 through 32 pseudobulks each gave 101 discoveries, while 33 gave 99. All 646 observed fits converged. The median number of valid permutations was 500; the minimum was 476 after excluding failed permutation refits. Asymptotic Dirichlet–multinomial p-values were anti-conservative, with 15.5% of permutation-null values below 0.05, which is why only exact per-event permutation ranks are used for the primary result.

Two sensitivity analyses did not improve the event-level result. Testing 3,708 individual cell-type pairs gave 97 FDR-significant pair hypotheses but only 32 distinct partitions after correction across all pairs. Restricting each block to its two molecule-weighted dominant paths increased coverage; with a 1% path floor it gave 88 discoveries, and removing the floor gave 91, under a cross-fitted empirical null. These analyses support the full-path exact omnibus test as the primary cell-type analysis.

Allowing the alternative to re-estimate κ also left the result unchanged: it found the same 101 partitions in the same 95 genes, with no event entering or leaving the 5% FDR set. Exact p-values from fixed- and free-concentration fits had Spearman correlation 0.9995, and both had 148 nominal events. The null optimum was multinomial for 499 of 646 tests, and the alternative optimum was multinomial for 580. Among 66 tests with finite concentration in both models, the median alternative/null concentration ratio was 1.54, consistent with cell-type mean effects explaining some apparent overdispersion. Exact permutation calibration absorbs this nuisance-parameter flexibility, so freeing κ provides no net power gain here.

Calibration

Across the 25-UMI analysis, 2.7% of permuted-label Wald p-values were below 0.05 for both conditional and profile covariance, indicating conservative null calibration. A parametric bootstrap of 3,000 reliable local fits gave mean squared standardized error 1.10 and 94.5% coverage for nominal 95% intervals. Independent Gaussian simulations of REML–GLS gave type-I error between 2.7% and 5.0% across the tested biological-variance settings.

Numerical tests additionally verify skipped-exon block construction, identifiable covariance after collapsing indistinguishable isoforms, rejection of information-null path contrasts, the analytic binomial Fisher variance, and recovery of imposed GLS effects.

Coverage-cutoff sensitivity

UMIs	blocks	tests	conditional null $p < .05$	profile null $p < .05$	bootstrap coverage
25	10100	1335	0.027	0.027	0.945
50	5896	755	0.031	0.030	0.936
100	2820	283	0.029	0.025	0.946
200	1034	69	0.028	0.026	0.948
500	195	10	0.030	0.050	0.946

No cutoff produced an FDR-significant block. The 500-UMI calibration estimate is based on only 100 permutation tests.

For reliable two-condition, two-path conditional tests, mean asymptotic power for a 0.5 ILR effect was 0.49, 0.51, 0.44, 0.56, and 0.59 at increasing cutoffs. The final two values are selected from only eight and two eligible tests. Among 103 hypotheses shared by the 25- and 50-UMI analyses, power was 0.453 and 0.449. Lower coverage cutoffs therefore increase the number of testable blocks without degrading calibrated precision among shared tests. The 25-UMI cutoff is the default.

Does cell resolution increase condition-test power?

The cell-resolved likelihood was run on the 527 blocks testable in the 25-UMI primary analysis. All 41,242 block-by-pseudobulk fits converged; 21,624 passed the reliability rules, compared with 21,010 pseudobulk conditional estimates on the same blocks. This produced 836 condition tests, compared with 829 for pseudobulk fitting. No cell-resolved test passed 5% FDR. Condition-label permutations were conservative, with 2.49% of null p-values below 0.05.

An exact matched comparison used the same reliable samples for both methods in 807 block-by-cell-type tests. Cell resolution reduced the median local covariance trace by 1.6%, and the median path-response RMS difference was 0.033 ILR units. P-values were strongly concordant (Spearman rank correlation 0.958); 24 pseudobulk and 25 cell-resolved tests were nominally significant.

For the 259 matched two-condition, two-path tests, mean asymptotic power for a 0.5 ILR effect was 0.504 for both methods. The median ratio of cell-resolved to pseudobulk condition-coefficient standard error was 0.995. Retaining cell likelihood terms therefore did not materially increase power in this dataset. The aggregated EC counts and decoder baseline preserve nearly all relevant local information within a cell type, while residual between-mouse variation dominates the small reduction in inferential variance. Pseudobulk fitting remains the default; the cell-resolved method is available for datasets with stronger within-pseudobulk isoform heterogeneity.

Compositional and joint-test assessment

Collapsing equivalent isoform partitions reduced the 829 conditional block-by-cell-type rows to 505 nonredundant tests over 317 partitions. The median effective path count was 32 molecules, or 62% of gene UMIs; only 0.6% of estimates reached the gene-UMI cap.

On the same 505 tests, conditional GLS had 12 nominal results and the Dirichlet–multinomial had 18. Neither had a 5% FDR discovery. The Dirichlet–multinomial was conservatively calibrated: 3.76% of 10,100 permuted-label p-values were below 0.05. In contrast, pure multinomial regression gave 197 nominal and 155 FDR results, but 39.1% of its permutation p-values were below 0.05. UMI sampling alone therefore cannot represent the large between-mouse compositional variation. Output tables mark pure multinomial and naive shared-effect rows as invalid for inference while retaining them as calibration diagnostics.

ACAT across cell types produced 8 nominal GLS and 12 nominal Dirichlet–multinomial partitions among 317 tests, with no FDR discoveries. Combining distinct partitions within genes gave 7 and 11 nominal genes among 276 tests, again with no FDR discoveries.

Naive shared-effect pooling illustrates the pseudoreplication risk. It gave 43 nominal and one FDR Dirichlet–multinomial partition, but synchronized mouse permutations rejected 13.2% of nulls at 0.05. The apparent discovery had asymptotic $p = 6.4 \times 10^{-5}$ but permutation $p = 0.143$. Naive shared GLS was also inflated, with 8.4% null rejection.

The mouse-clustered shared GLS restored conservative calibration: 2.19% of 7,169 synchronized permutation p-values were below 0.05. It produced 11 nominal partitions among 359 tests and no FDR discoveries; its smallest p-value was 0.0126 with FDR 0.86. Gene-level ACAT gave 9 nominal genes among 313 and no FDR discoveries. Thus the calibrated gain is limited to the increase from 12 to 18 nominal within-cell-type tests under the Dirichlet–multinomial. None of the evaluated strategies creates a genome-wide condition discovery in this dataset.

Limitations

The profile covariance is conditional on the fitted genome-wide decoder. Uncertainty in shared decoder parameters is expected to be small for common paths but may matter for rare genes or cell types. A mouse-level jackknife of the hierarchical fit is the preferred extension.

The current biological variance is isotropic in log-ratio space. Estimating a full covariance separately for every block would be unstable with few mice. A later empirical-Bayes model could shrink path-specific biological covariances across genes.